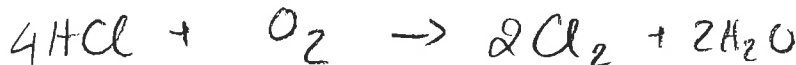


To receive full credit, you must show all your work.

- (1) (55 pts.) Hydrogen chloride (HCl) and oxygen (O₂) are reacted in the gas phase to form chlorine (Cl₂) and water (H₂O). A gas feed stream entering the reactor contains 100 mol/min of HCl and air (79 mol% N₂, 21 mol% O₂). The air in the stream contains 30% more O₂ than stoichiometrically required to oxidize all the HCl. In the reactor, 60% of the HCl is oxidized. The product stream contains all the Cl₂ and H₂O produced, the inert N₂, and the unreacted portion of the HCl and O₂.

- (a) Write a balanced chemical reaction equation for this reaction. Use any method you like to balance the equation. (5 pts.)



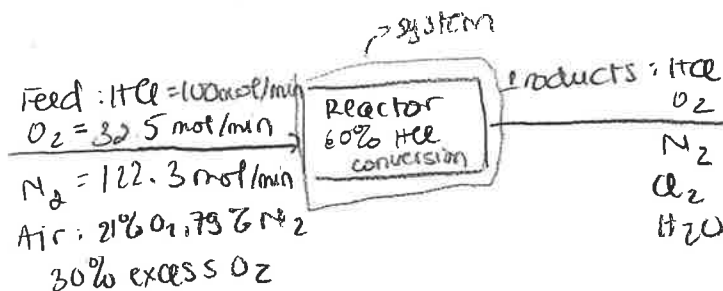
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⇒ total independent equations:

$$5a + 5b + 5c + 5d = 1 + 1 + 2 + 5 = 9$$

$$\Rightarrow \text{DOF} = 9 - 9 = 0$$

- (b) Draw and label a block flow diagram for the process. (5 pts.)



- (c) Do a degree of freedom analysis following the method on page 105 of the textbook by Murphy. Give values for 4a, 4b, 5a, 5b, 5c, and 5d. Compute the DOF. (5 pts.)

Choose components in system: HCl, O₂, N₂, Cl₂, H₂O

→ Product stream has 5 components → 5

$$4a = 3 + 5 = 8$$

4b. Independent reactions = 1
Accumulation = 0 (steady state)

$$\Rightarrow 4b = 1$$

$$\Rightarrow \text{Total independent} = 4a + 4b = 8 + 1 = 9$$

$$n_{\text{HCl, in}} = 100 \text{ mol/min} \rightarrow 5a = 1$$

• Air composition: 21% O₂, 79% N₂ → 5b = 1
components, only 1 independent spec

• 30% excess O₂ relative to stoich for complete HCl oxidation → 1

• 60% HCl oxidized → 1

$$\Rightarrow 5c = 2$$

• number of components in system = 5 → 5d = 5

- (d) Calculate the molar flow rates of the O₂ and N₂ in the feed stream to the reactor. (10 pts.)

Stoichiometric O₂ needed to oxidize all HCl:

from: 4HCl : 1O₂

⇒ for 100 mol/min HCl:

$$\dot{n}_{\text{O}_2, \text{stoch}} = 100 \left(\frac{1}{4} \right) = 25 \text{ mol/min}$$

$$\dot{n}_{\text{O}_2, \text{in}} = 1.3 \times 25 = 32.5 \text{ mol/min}$$

Air has 21% O₂:

$$\dot{n}_{\text{air}} = \frac{32.5}{0.21} = 154.7619 \text{ mol/min}$$

Then N₂ is 79%:

$$\dot{n}_{\text{N}_2, \text{in}} = 0.79 \times 154.7619 = 122.2619 \text{ mol/min}$$

Final answer

$$\Rightarrow \text{O}_2 = 32.5 \text{ mol/min}$$

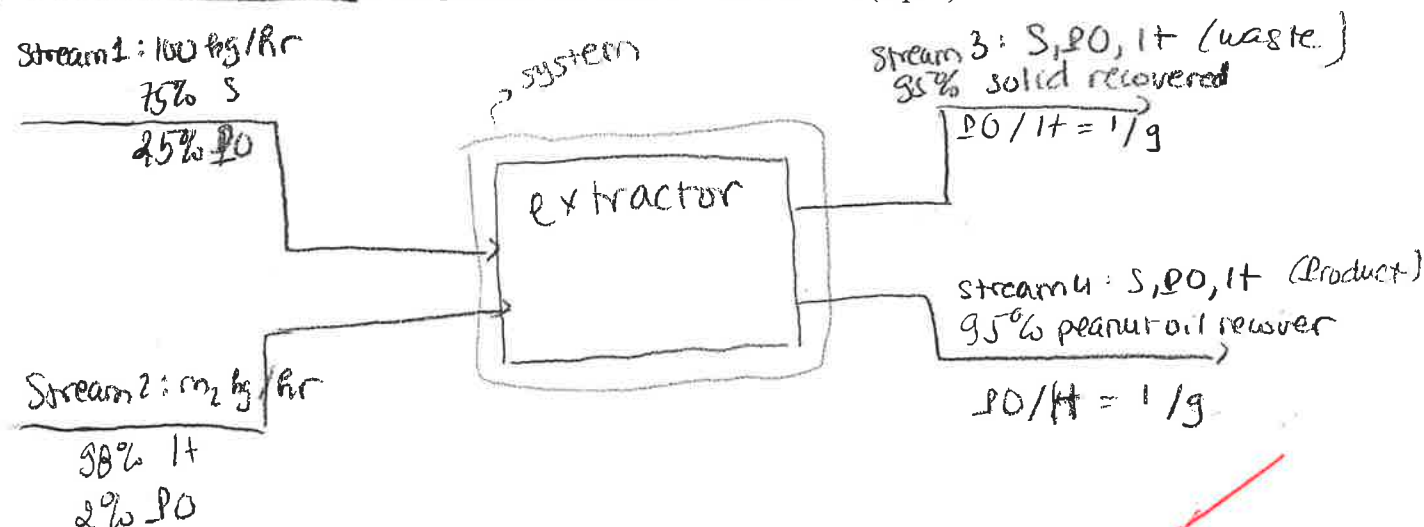
$$\Rightarrow \text{N}_2 = 122.26 \text{ mol/min}$$

Feed Stream molar flow rate:

Acid (HCl) (mol/min)	Oxygen (O ₂) (mol/min)	Nitrogen (N ₂) (mol/min)
100 mol/min	32.5 mol/min	122.26 mol/min

(2) (45 pts.) Hexane can be used to extract peanut oil from ground peanuts. There are two streams entering the extractor unit: (1) the feed stream $\dot{m}_1 = 100 \text{ kg/hr}$ containing ground peanuts which has a composition of 75 wt% solids (S) and 25 wt% peanut oil (PO), and (2) a stream $\dot{m}_2$ of 98 wt% hexane (H) and 2 wt% peanut oil. There are two streams exiting the extractor unit: (3) a waste stream $\dot{m}_3$ containing solids, peanut oil and hexane, and (4) the product stream $\dot{m}_4$ which also contains solids, peanut oil and hexane. The waste stream recovers 95% of the solids entering the extractor while the product stream recovers 95% of the peanut oil entering the extractor. The equilibrium solubility of peanut oil in hexane is 1 kg of peanut oil to 9 kg of hexane. Assume that this equilibrium has been achieved in both exit streams.

(a) Draw a block flow diagram of the process and label all the streams. (5 pts.)



(b) Solve the appropriate mass balances along with the recovery equation above to calculate all the species and total mass flow rates in kg/hr. Fill in the table below with your answers. (40 pts.)

Stream 1: total = 100 kg/hr
 $S_1 = 75$, $PO_1 = 25$, $H_1 = 0$ ✓

Stream 2: $PO_2 = 0.02m_2$
 $H_2 = 0.98m_2$
 $S_2 = 0$ ✓

All solids come from stream 1 = 75 kg/hr ✓

Waste gets 95%: $S_3 = 0.95(75) = 71.25$ ✓

Remaining to stream 4: $S_4 = 75 - 71.25 = 3.75$

Oil entering extractor:

$PO_{in} = 25 + 0.02m_2$

Product gets 95%: $PO_4 = 0.95(25 + 0.02m_2)$ ✓

(Problem 1 Continued)NAME Tran Ngoc Trinh

(e) Calculate the species molar flow rates for all five components in the product stream in mol/min. Show all your calculations and write your final answers in the space provided below. (30 pts.)

$$\text{HCl reacted} : 0.6 \times 100 = 60 \text{ mol/min}$$

$$\text{unreacted HCl} : 100 - 60 = 40 \text{ mol/min}$$

From stoichiometry: 4 mol HCl require 1 mol O_2

$$\Rightarrow \text{O}_2 \text{ consumed} = 60 \times \frac{1}{4} = 15 \text{ mol/min}$$

$$\Rightarrow \text{O}_2 \text{ remaining} : 32.5 - 15 = 17.5 \text{ mol/min}$$

From reaction: $4\text{HCl} \rightarrow 2\text{Cl}_2$

$$\Rightarrow \text{Cl}_2 = 60 \times \frac{2}{4} = 30 \text{ mol/min}$$

From reaction: $4\text{HCl} \rightarrow 2\text{H}_2\text{O}$

$$\Rightarrow \text{H}_2\text{O} = 60 \times \frac{2}{4} = 30 \text{ mol/min}$$

$$\text{Nitrogen is inert} \Rightarrow \text{N}_{2,\text{out}} = 122.26 \text{ mol/min}$$

30

$$\begin{array}{r|l} 1 & 55 \\ \hline 2 & 45 \\ \hline T & 100 \end{array}$$

Product Stream molar flow rate:

Acid (HCl) (mol/min)	Oxygen (O_2) (mol/min)	Nitrogen (N_2) (mol/min)	Chlorine (Cl_2) (mol/min)	Water (H_2O) (mol/min)
40	17.5	122.26	30	30

55

(Problem 2 Continued)

waste gets 5% : $PO_3 = 0.05(25 + 0.02m_2)$

For both exit streams: $\frac{PO}{H} = \frac{1}{9}$ ✓

$$\Rightarrow H_3 = 9PO_3$$

$$\Rightarrow H_4 = 9PO_4$$

Hexane in : $0.98m_2$

Hexane out : $H_3 + H_4$

$$\text{Sub : } 0.98m_2 = 9PO_3 + 9PO_4$$

$$\text{But : } PO_3 + PO_4 = (25 + 0.02m_2)$$

$$\Rightarrow 0.98m_2 = 9(25 + 0.02m_2)$$

$$\Rightarrow 0.98m_2 = 225 + 0.18m_2$$

$$\Rightarrow 0.8m_2 = 225 \Rightarrow m_2 = 281.25 \text{ kg/hr}$$

$$\text{Stream 2 : } PO_2 = 5.625$$

$$H_2 = 275.625$$

$$\Rightarrow \text{total} = 281.25$$
 ✓

$$\text{total oil in : } 25 + 5.625 = 30.625$$

$$\text{Stream 4 oil : } PO_4 = 0.95(30.625) = 29.094$$

$$\text{Stream 3 oil : } PO_3 = 1.531$$

$$\text{Hexane amounts : } H_4 = 9(29.094) = 261.844$$

$$H_3 = 9(1.531) = 13.781$$

40

45

Species: Stream#	1	2	3	4
PO (kg PO/hr)	25	5.625	1.531	29.094
S (kg S/hr)	75 ✓	0	71.25 ✓	3.75 ✓
H (kg H/hr)	0	275.625	13.781	261.844
Total (kg/hr)	100	281.25	86.562	294.688